Learning Action Priors for Fog of War Chess

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Introduction

Can a learned action prior identify strategically relevant and information-gathering actions, enabling more efficient search in Fog of War Chess while preserving playing strength?

Fog War (FoW) Chess is a Chess variant where each player can only see the board tiles where their pieces are able to move to (visualized below). Due to this incomplete-information aspect dominating the game, creating an agent to play FoW Chess is more similar to solving poker than normal chess. Our project focuses on learning an action prior for decision making under uncertainty. Unlike PPO, whose objective is to directly select the final action, the action prior identifies which actions deserve further computation. Similar to how a retriever selects relevant documents before an LLM performs deeper reasoning, the action prior selects a small set of promising candidate moves before search is applied.

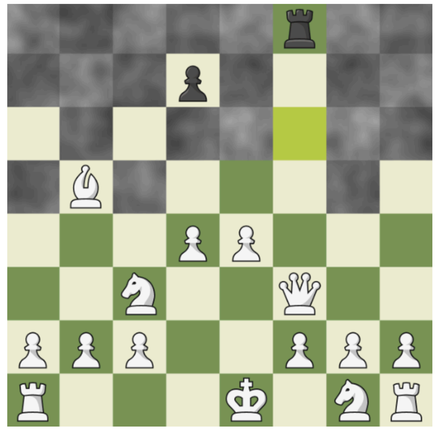


Figure 1. Fog Of War Board

Why Action Priors Matter

- **Partial Observability**
  The true board state is unknown, requiring the agent to reason over multiple plausible opponent configurations.
- **Large Effective Search Space**
  - Typical position: **30–40 legal moves**
  - Depth 4: $\mathbf{35^4 \approx 1.5M}$ move sequences
  - Depth 6: $\mathbf{35^6 \approx 1.8B}$ move sequences
- **Information-Gathering Actions**
  Unlike standard chess, seemingly weak moves may be valuable because they reveal hidden pieces and reduce uncertainty.

Key Idea: Existing pruning methods prioritize immediate tactical value. Our learned action prior prioritizes both strategically relevant and information-gathering actions, enabling more efficient search under partial observability.

Methods

Overall Approach



Key Distinction: PPO learns *which move to play*, while the action prior learns *which moves deserve further computation*.

1. State Representation

The FoW board state is encoded as a $\mathbf{14 \times 8 \times 8}$ tensor comprising piece-type occupancy maps, visible enemy piece maps, square visibility, and side-to-move features.

2. Action Prior Network

We learn an action prior $\pi_{prior}(a|o)$ that maps partial observations to action scores, inducing a reduced candidate set $A_k \subseteq A$ for search.

3. Search Procedure

Lookahead search is performed only over $A_k \subseteq A$, enabling deeper search under a fixed computational budget while preserving high-value candidate actions.

4. Training

Policies are optimized via PPO self-play with dense reward shaping (+1 win, -1 loss, capture/check rewards). Search depth and candidate set size (k) are fixed across all experiments.

Baselines

Method	Description
Dense PPO	Policy-only agent
Policy Prior + Search	AlphaZero-style policy prior
Heuristic Prior + Search	Hand-designed pruning
Information Prior + Search (Ours)	Learned information-aware prior

5. Evaluation

- Round-robin self-play tournament
- Metrics: **Win Rate** and **Search Nodes Evaluated**

Results

Primary Result: Win Rate Comparison

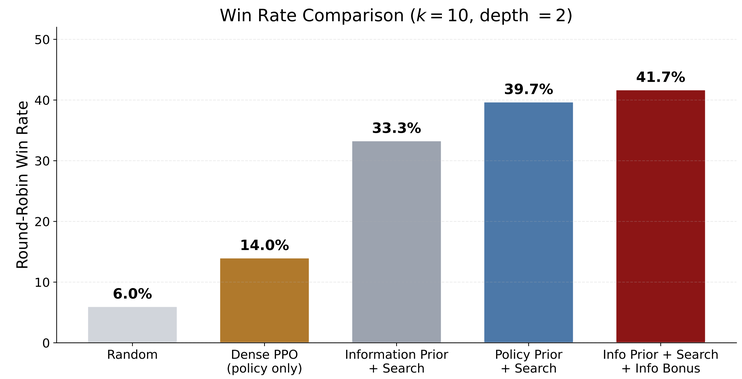


Figure 2. Round-robin win rate (k=10, depth =2, 100 games/direction).

Key Findings

- Information Prior + Search: **41.7%** win rate
- Relative improvement: **+2.0 pp** over policy prior
- Search is decisive: policy-only PPO (14%) $\ll$ prior + search ($\sim$40%)

Discussion & Significance

Top-k Ablation

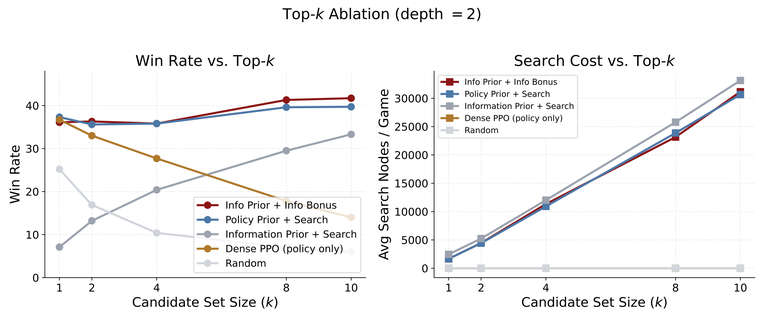


Figure 3. Win rate and search cost vs. candidate set size k.

Takeaway

- Action priors enable efficient restricted search under partial observability.
- Adding FoW information gain at *selection time* (top-k) outperforms a plain policy prior and is more robust across k.
- Baking information reward into training alone under-ranks tactical moves at small k; inference-time info avoids this miscalibration.